

ZHENGYAN HU

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Software engineer specializing in LLM agent infrastructure — agent runtime, evaluation, and the reliability layers that make agentic systems production-ready. Backed by a strong distributed-systems and backend foundation.

EXPERIENCE

Intuit (via Pias Technology, Contract)

San Jose, CA

Software Engineer — GenAI org, LLM Agent Infrastructure

04/2026 – Present

- Migrated a legacy **multi-agent architecture** to a modular, **skill-based agent framework**, decomposing monolithic agent logic into composable, independently testable skill components to improve maintainability and scalability.
- Built an **LLM-as-a-judge evaluation framework** scoring agent output on accuracy, relevance, completeness, and coherence, enabling automated quality-regression checks between releases and a confident, incremental migration cutover.
- Integrated **agent identity authentication** into backend services, exchanging user tokens for scoped service identities via an auth SDK, with a **feature-flag-controlled rollout** and fallback logging for safe production migration.
- Decoupled the **service-to-service client for agent orchestration** from the main request flow, introducing a clean interface boundary that improved modularity and testability.
- Hardened the release path: added an integration-test stage to the **CI/CD pipeline (Jenkins / Argo CD)**, and built **Splunk** alerting and log analysis to detect anomalies, trace root causes, and resolve production incidents end-to-end.

Avary.ai

San Jose, CA

Software Engineer

08/2025 – 04/2026

- Owned the full lifecycle of a cloud-native **AI evaluation pipeline (Spring Boot, OpenAI API)** serving 20+ enterprise clients — from design through deployment and on-call operation.
- Reduced latency **50%** and lowered API cost via **Redis caching and parallel multi-prompt inference** (Java CompletableFuture).
- Designed a **fault-tolerant asynchronous pipeline** with AWS SQS (3k+ messages/day) and DLQ-based retry logic, achieving a **99.5% success rate** with <2s p95 latency under peak load.
- Developed **RBAC-based access control** for sensitive candidate data, enforcing role-based permissions across backend APIs.

Baylor University

Waco, TX

Software Researcher — Distributed Systems / Microservices

01/2023 – 05/2023

- Built an automated detector for architectural anti-patterns in distributed microservice systems, using **DFS-based graph algorithms** to identify cyclic dependencies and bottlenecks.
- Architected a real-time telemetry pipeline (Python/Flask) ingesting **Jaeger trace data** to visualize interactions across 37+ microservices.

PROJECTS

Paxos Distributed Key-Value Store

09/2024 – 12/2024

- Implemented a fault-tolerant key-value store using the **Paxos consensus algorithm** (Java RMI), maintaining strong consistency across 5 replicas; stress-tested a Dockerized multi-node cluster with 500+ concurrent requests under simulated partitions and node failures.

Cloud-Based S3 / Object-Storage Monitoring System

09/2024 – 12/2024

- Architected a serverless monitoring service using SNS/SQS fan-out and **AWS Lambda** for real-time S3 events; designed **DynamoDB GSIs** for sub-50ms analytics queries, with CloudWatch-triggered automated remediation.

SKILLS

AI / GenAI: LLM agents, agent runtime & orchestration, LLM-as-a-judge evaluation, prompt engineering, RAG, OpenAI API

Languages: Python, Java, Go, SQL, JavaScript/TypeScript, C/C++

Backend & Systems: Spring Boot, REST APIs, microservices, distributed systems, OOD, data structures & algorithms, JUnit, Git

Cloud & DevOps: AWS (Lambda, S3, SQS/SNS, DynamoDB, CloudWatch, CDK), Docker, CI/CD, Jenkins, Argo CD, Redis, Prometheus

EDUCATION

Northeastern University — M.S. Computer Science, GPA 4.0/4.0

09/2023 – 05/2025

Baylor University — B.S. Computer Science

09/2019 – 05/2023